

SAFE WORK METHOD STATEMENT (SWMS)

In order of priority:

- 1. ELIMINATION** – Remove hazard completely.
- 2. Substitution** – Replace object /process with less hazardous one.
- 3. Isolation** – Isolate the hazard (i.e. separate the hazard from the people).
- 4. Engineering** – Modify a process (i.e. redesign a workstation).
- 5. Administration** – Use procedures /systems to minimise the risk (i.e. Rotation of employees, implementation of a safe work method, training, etc).
- 6. Personal Protective Equipment (PPE)** – use to try to protect the person.

LIKELIHOOD	RISK SCORE CALCULATOR				
Certain to Occur	Mod.	High	High	Extreme	Extreme
Very Likely	Mod.	Mod.	High	Very High	Extreme
Possible	Low	Mod.	High	Very High	Very High
Unlikely	Low	Mod.	Mod.	High	Very High
Rare	Low	Low	Low	Mod.	High
CONSEQUENCE	LOW	MINOR	MODERATE	FATAL	CATASTROPHIC

Residual Risk	Action Required
Low	Ok for now. Continue to monitor.
MODERATE	Determine if remedial action/treatment is required.
HIGH	Immediate decision required by site leadership. Identify and commence remedial action with appropriate priority.
VERY HIGH	ACT NOW. Requires immediate attention by site leadership. Address the risk without delay. Requires urgent attention and clear, timely action plans to reduce the risk before proceeding.
EXTREME	ACT NOW. Requires immediate attention by site leadership. Urgently address the risk. Requires immediate resolution with focused attention until resolved.

Identify task high risk workplace hazards - *Non routine work is work not documented in a Safe Operating Procedure (SOP)* or Work Instruction

Note: If machine isolation is required, workers must be competent in the relevant isolation procedure.

EXTREME & VERY HIGH RISK HAZARDS	Risk	MAIN RISKS / PERMITS REQUIRED
1. Entering a Confined Space.	E	- Heat exhaustion, engulfment, limited access/egress. - Asphyxiation – low O₂ or contaminants. (Confined Space Entry Permit required) - Explosive atmosphere, combustible dusts (Confined Space Entry/Hot Work Permits required)
2. Work in an area which may have a contaminated or flammable atmosphere.	E	- Fire / explosion from ignition source. (Hot Work Permit required) - Asphyxiation or poisoning from contaminated atmosphere. (Confined Space Permit)
3. Work with potential for a fall from 2m or work on a roof	E	Falling worker/object ≥2m requires SWMS
4. Hot work required. (Oxy cutting, welding, soldering, grinding, etc).	E	- Fire or explosion from ignition source. (Hot Work Permit required) - Injury caused by hot surface, grinding wheel ejected pieces, welders "flash", etc. Fire inside Expanded Polystyrene (EPS) wall panels (Hot Work Permit required).
5. Hand dig >200mm OR ground penetration >200mm OR mechanical earthworks OR enter a trench >1.5 M deep (excavating is Construction work)	E	- Contact utilities – electricity, water, gas, communications. (Excavation Permit required) - Trench Collapse. (Excavation Permit required)
6. Non routine electrical, high voltage or 'live' work or work near electrical installations.	E	- High voltage is >33kV. 'Live' work is when electrical equipment is worked on energised – (Management Approval Required) - Electrical cabinets or MCC identified as Red
7. Non routine work involving mechanical lifting, lowering, pushing or pulling.	E	- High mechanical force damage / injury. - High risk work licence for use of cranes, mobile plant, scaffold, dogging, etc.
8. Using high risk tools with excess noise, kickback, cutting edge, high speed/pressure, ejected material, weight, awkward posture, explosive force.	E	- Injury from use of high risk powered or hand tools. - Electrocution, burn or electric shock. - Manual task injury from weight, forces, overexertion, repetition or poor work heights.
9. Demolish / alter load bearing / integral part of a structure 6m high OR use load shifting machinery on a suspended floor OR use explosives. Includes modify handrail / floors.	E	Being struck or crushed by parts of a structure or exposure to asbestos, lead, hazardous chemicals etc. (Demolition License and/or asbestos removal licence and Safe Work Plan required)
10. Work inside fences / barricades / enclosures where moving / rotating machinery is operating or hazards may exist.	V	- Serious bodily injury caused by contacting moving/unguarded plant or materials. - Machinery starting unexpectedly or electrocution, burn or electric shock.
11. Potential or stored energy which cannot be isolated (electrical, hydraulic, pneumatic, kinetic).	V	- Electrocution or entrapment. - Machinery entanglement / injury.
12. Work on or near a pressurized or enclosed circuit (water, pneumatic, hydraulic, hot oil, etc).	V	- Fracturing the mains or piping from contact – spills / leaks / emissions (environment). - Contacting temperature extremes.
13. Working in artificial temperature extremes or weather extremes. Hot, cold, wet, icy, lightning, high winds.	V	- Electrocution from using power tools / leads in wet conditions or lightning. - Heat stroke / Heat Exhaustion / Fatigue / Frost bite / Hypothermia / Influenza. - Slip / Trip / Fall Hazards.
14. Hazardous manual tasks or excessive noise or excessive vibration.	V	Personal injury/illness or hearing loss.
15. Using a hazardous chemical – chemicals, heavy metals, asbestos, Lead.	V	- Exposure to hazardous chemicals. - Spills / leaks (environmental).

CONTROLLED DOCUMENT

Doc Id: SWMS Template	Page 1 of 5
Doc Owner: Tuan Safety and Compliance Officer	Issue Date: 20 November-2024
Review: 20-Nov-2024	Version No: 001

Key Steps in Logical Order		Hazards/Risks to Workers or Public	Risk Score	Control Measures	Residual Risk Score
1	Isolate Plant and Equipment and prepare the work area (clean out surge bin ready for hot work)	- EWP and Cranes, - Accidental Machine Startups, - Falling objects, - Confined space - respiratory irritation or eye injuries from residual dust and airborne particles	High	- Isolate the New York Blower, Surge bin auger 1 and 2, Surge bin airlock. - The area below overhead work shall be marked out and an exclusion zone setup to prevent unauthorised entry. - Open all inspection hatches - Confined space permit. Including standby person with constant radio contact. - Respiratory protection and safety goggles or glasses to be worn.	Mod
2	Disconnect the Grecon and Start removing the ducting	- Working at heights - Pinch points, - Slips, Trips, - Falling objects - Working at heights - structural instability	Very High	- Competent crane driver and dogman. - House-keeping and extra care. - PPE to be used (gloves and hard hat) - inspect all connections and supports for stability	Mod
2	Start Removing Cyclones and support Frame	- Working at heights - Pinch points, - Slips, Trips, - Falling objects - Fire from Hot Work, - Hazards related to welding/grinding	Very High	- All work to be carried out by competent person/s - The area below overhead work shall be marked out and an exclusion zone setup to prevent unauthorised entry. - Hot works permit (Including stand-by person and fire watch person). -Use a spotter when required. -Simultaneous work on multiple levels is not permitted unless suitable monitoring provisions and safety barriers are used.	Mod
4	Cut and Remove Surge Bin Lid half and place on ground.	- Working at heights - Pinch points, - Slips, Trips, - Falling objects - Fire from Hot Work, - Hazards related to welding/grinding	Very High	- All work to be carried out by competent person/s - The area below overhead work shall be marked out and an exclusion zone setup to prevent unauthorised entry. - Hot works permit (Including stand-by person and fire watch person). -Use a spotter when required. -Simultaneous work on multiple levels is not permitted unless suitable monitoring provisions and safety barriers are used.	Mod

CONTROLLED DOCUMENT
Doc Id: SWMS Template

Page **3** of **5**
Doc Owner: Tuan Safety and Compliance Officer

Issue Date: 20 November-2024

Review: 20-Nov-2024

Version No: 001

SAFE WORK METHOD STATEMENT (SWMS)

5	Cut end wall out of Surge Bin lid on ground and install back up onto surge bin	- Working at heights - Pinch points, - Slips, Trips, - Falling objects - Fire from Hot Work, - Hazards related to welding/grinding	Very High	- Only work behind guardrails unless fall protection equipment is in use. - All work to be carried out by competent person/s. - The area below overhead work shall be marked out and an exclusion zone setup to prevent unauthorised entry. - Hot works permit (Including stand-by person and fire watch person). -Use a spotter when required.	Mod
6	Assemble dust collector, hopper, frame and platforms on the ground.	- Manual handling, - Working with heavy equipment - pinch points	Mod	-Mechanical aid forklifts or crane - Ensure proper use of lifting equipment and inspect before use - Perform team lifts -Clear Communication between workers positioning parts.	Low
7	Install support beams and bin connection for the dust collector.	- Working at heights - Ewp and crane - Manual handling - Falling objects - Pinch points and moving parts	Very High	- Only work behind guardrails unless fall protection equipment is in use - Competent crane driver and dogman. - The area below overhead work shall be marked out and an exclusion zone setup to prevent unauthorised entry. - Communicate clearly during component positioning -Simultaneous work on multiple levels is not permitted unless suitable monitoring provisions and safety barriers are used.	Mod
8	Install dust collector assembly.	- Working at heights - Ewp and crane - Manual handling - Falling objects - Pinch points and moving parts	Very High	- Only work behind guardrails unless fall protection equipment is in use - Competent crane driver and dogman. - The area below overhead work shall be marked out and an exclusion zone setup to prevent unauthorised entry. - Communicate clearly during component positioning	Mod

CONTROLLED DOCUMENT

Doc Id: SWMS Template

Page **4** of **5**

Doc Owner: Tuan Safety and Compliance Officer

Issue Date: 20 November-2024

Review: 20-Nov-2024

Version No: 001

SAFE WORK METHOD STATEMENT (SWMS)

9.	Install ducting from the hammer mill to the dust collector	- Working at heights - Ewp and crane - Manual handling - Falling objects - Pinch points and moving parts - structural instability	Very High	- EWP with fall protection equipment is in use - Competent crane driver and dogman. - Communicate clearly during component positioning - inspect all connections and supports for stability - Simultaneous work on multiple levels is not permitted unless suitable monitoring provisions and safety barriers are used.	Mod
10.	Install ducting from the fan to the dust collector	- Working at heights - Ewp and crane - Manual handling - Falling objects - Pinch points and moving parts - structural instability	Very High	- EWP with fall protection equipment is in use - Competent crane driver and dogman. - Communicate clearly during component positioning - inspect all connections and supports for stability	Mod

Attach additional pages if required

All relevant workers to be consulted during this process and made aware of the control measures and all required permits identified.

AUTHORISATION			
Approved by	David Knight	Position	Plant manager
Signed		Date	

Name, Sign Off & Employer of Persons Performing Work		Start Time:	am/pm	Finish Time:	am/pm
Name	Sign Off	Employer			
1					
2					
3					
4					
5					

JOB COMPLETION					
1. Has equipment been modified/changed to affect how it operates?		☐ N (go to 3) ☐ Y (sign off by competent operator required)			
Has a <u>competent operator</u> approved it is safe to use? ☐ Y ☐ N		Competent Operator Name:	Competent Operator Signature:		
2. Work completed & area assessed safe before return to normal service? ☐ Y ☐ N <i>To be completed by the person conducting the task</i>			Name:		
			Signature:		
3. Altus Representative advised of & accepts changes: ☐ Y ☐ N		Name:		Signature:	

CONTROLLED DOCUMENT			
Doc Id: SWMS Template	Page 5 of 5		
Doc Owner: Tuan Safety and Compliance Officer	Issue Date: 20 November-2024		
Review: 20-Nov-2024	Version No: 001		